The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2

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Abstract. The algebraic R-flux algebra — the non-Lie Malcev algebra obtained from the imaginary octonions by Inönü–Wigner contraction, whose central element W is proven representable at every degree in its non-associative envelope — cannot be represented by commutators in any associative algebra with $W \neq 0$: the Jacobi identity forces $W = 0$, and by the locked normalization established earlier in this series, the canonical pairing $\{X, P\} \sim W$ dies with it. We show this obstruction is evaded not by representation but by a *second* Inönü–Wigner contraction, with weights $(1, 1, 2)$ on (X, P, W): the R-flux bracket and the Jacobiator die linearly in the contraction parameter while the Heisenberg bracket survives bit-identically, landing on the $3+3+1$ Heisenberg Lie algebra with W alive and central. This identifies the two automorphism orbits of valid octonion contractions, classified in the companion papers as "genuine" versus "trivial," as two stations of one tower. We further prove three facts about the grading dilation $(\lambda, \lambda^2, \lambda^3)$: it is not a symmetry of the parent octonions; it is created exactly at the first contraction and is an automorphism of every member of the second-contraction family; and on the fixed-W (Weyl) algebra it fails by exactly λ^3, splitting into a W-preserving squeeze flow and a W-rescaling component. Finally, we compute the subgroup of G_2 preserving the grading: the chain is $G_2\ (14) \supset \mathfrak{su}(3)\ (8) \supset \mathfrak{so}(3)\ (3)$, with the surviving $\mathfrak{so}(3)$ acting as one and the same rotation on the X-triple and the P-triple and surviving the entire tower as automorphisms. Within this model, the isotropy of the three-dimensional position and momentum spaces is derived, not postulated. A closing section states, with citations only, the operator-algebraic program these results prepare.

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1. Introduction

Earlier papers in this series establish the algebraic R-flux algebra as a genuine non-Lie Malcev algebra obtained from $\mathrm{Im}(\mathbb{O})$ by a graded Inönü–Wigner contraction (weights $1, 2, 3$ on generators X, P, W), with W centrally embedded at every degree of its Pérez-Izquierdo–Shestakov envelope and with the canonical pairing, the central value, and the Jacobiator locked to a single parameter (*The Central Element of Algebraic R-Flux*). Separately, the family of such contractions was classified: 56 of 140 gradings are structurally valid, splitting into two automorphism orbits of 28 — "genuine" (momenta close; non-associative R-flux survives) and "trivial" (coordinates close; only a Heisenberg algebra remains) (*Which Contractions of the Octonions Give Genuine R-Flux*; *Malcev-Preservation Is Generic, Not Exceptional*).

This note answers three questions those papers left open, and which any attempt to connect the non-associative stratum to ordinary quantum mechanics must face. First: since associativity provably kills W (§2), how does canonical quantum mechanics — associative, with $\hbar \neq 0$ — relate to this structure at all? Second: the orbit counts $(28 + 28)$ were verified but never explained; what is their origin? Third: what symmetry, if any, survives from $\mathrm{Aut}(\mathbb{O}) = G_2$ once a grading is fixed — and hence constrains any state built downstream?

2. The obstruction: associativity forces $W = 0$, and the lock makes it total

Proposition 1. Let M be the algebraic R-flux Malcev algebra, with brackets $\{X^a, X^b\} = 2\epsilon^{abc} P_c$, $\{X^a, P_b\} = -2\delta^a_b W$, $\{P_a, P_b\} = 0$, W central, and Jacobiator $J(X^1, X^2, X^3) = \pm 12\, W$ (orientation-dependent sign; magnitude 12 verified directly). Let $\rho:M\to A$ be a linear map into an associative algebra A satisfying $\rho(\{x, y\}) = [\rho(x), \rho(y)]$ for all $x, y \in M$. Then $\rho(W) = 0$, and consequently $\rho(\{X^a, P_b\}) = 0$ for all a, b.

Proof. Commutators in an associative algebra satisfy the Jacobi identity, so ρ applied to any Jacobiator vanishes. Applied to $J(X^1, X^2, X^3) = \pm 12\, W$: $\rho(W) = 0$. The second claim is the locked normalization: $\{X^a, P_b\} = -2\delta^a_b W$, so $\rho(\{X^a, P_b\}) = -2\delta^a_b\, \rho(W) = 0$. ■

The force of Proposition 1 comes from the lock. Killing the Jacobiator alone would be "flux-classicality"; but because the same W, at fixed ratio, sits in both the Jacobiator and the canonical pairing, an associative bracket-representation of the *full* algebra is fully classical — no uncertainty relation survives. Associativity cannot be imposed on this algebra at any nonzero $\hbar$. (This does not conflict with the Pérez-Izquierdo–Shestakov envelope, which is non-associative by design; it delimits what *associative* machinery can carry.)

3. The incidence structure of genuine gradings: $28 = 7 \times 4$, and the mediator property

The classification papers verified, but did not explain, the orbit count 28. Both the count and two structural properties follow from pure Fano-plane incidence.

Fix a Fano line P (the closed triple) and a point $w \notin P$ (the mediator, W's direction); let X be the remaining three points.

Proposition 2 (validity is automatic). Every such pair (P, w) yields a Weimar-Woods-valid grading. *Proof.* The unique non-trivial validity constraint is that no product of two X-points lands on w. The three lines through w partition the remaining six points into a perfect matching (the matching lemma of *The X/P Asymmetry Was Never a Law*, §2). Each line through w meets the line P in exactly one point (two distinct Fano lines meet in exactly one point, and $w \notin P$); hence each matched pair contains exactly one P-point and one X-point. No pair with both members in X exists, so no X-product reaches w. ■

Proposition 3 (the mediator property). $[w, P] \subseteq X$: multiplication by the mediator carries the closed triple into the open triple. *Proof.* For $p \in P$, the third point of the line through w and p is neither in P (that line meets P only at p) nor equal to w; hence it lies in X. ■

Corollary (the count). Genuine gradings are exactly the pairs (Fano line, external point): $7 \times 4 = 28$ — precisely Orbit B of the classification. All three statements were verified computationally: all 28 pairs are valid, genuine, and satisfy the mediator property, with zero exceptions. In the standard Cayley–Dickson labeling the mediation is exact and sign-free: $e_i\, e_4 = e_{i+4}$ for $i = 1, 2, 3$, with the co-triple's internal products landing back in the closed triple.

4. The second contraction

Definition. For $\mu \in [0, 1]$, let M_μ have brackets $\{X^a, X^b\} = 2\mu\, \epsilon^{abc} P_c$, $\{X^a, P_b\} = -2\delta^a_b W$, $\{P, P\} = 0$, W central. This is the Inönü-Wigner contraction of $M = M_1$ along the weights $(1, 1, 2)$ on (X, P, W): with $X_\mu = \mu X, P_\mu = \mu P, W_\mu = \mu^2 W$, the pairing $\{X_\mu, P_\mu\} = -2\delta\, W_\mu$ is exactly μ-independent, while $\{X_\mu, X_\mu\} = 2\mu\, \epsilon P_\mu$.

Theorem 1. (i) The R-flux bracket and the Jacobiator vanish linearly in μ, while the Heisenberg bracket is exactly constant: verified numerically, $|\{X^1, X^2\}| = 2\mu$ and $|J(X^1, X^2, X^3)| = 12\mu$ at $\mu = 1,\ 0.1,\ 0.01,\ 0$, with the W-coefficient of $\{X^1, P_1\}$ equal to -2.0000, bit-identical, at every μ. (ii) M_0 satisfies the Jacobi identity on all 35 basis triples: it is the $3{+}3{+}1$ Heisenberg Lie algebra, with W alive and central. (iii) The obstruction of §2 is thereby evaded, not violated: the contraction changes the algebra, so Proposition 1 — which concerns representations of the *full* algebra — no longer applies; M_0 has identically vanishing Jacobiator, and its associative representation theory (Stone–von Neumann) carries $W = \hbar \cdot 1 \neq 0$.

The mechanism deserves emphasis. The lock of §2 is a property of the octonionic level; the $(1, 1, 2)$ weights scale the Jacobiator-to-pairing ratio to zero *structurally*, without ever tuning W. What is shed in the passage to associativity is exactly and only the R-flux content; what survives is canonical quantum mechanics, $\hbar$ included — with W not transported as an independent generator but *regenerated on arrival* as $W = -\frac{1}{2}\{X^1, P_1\}$, the mediator recovered as the relation between the surviving triples.

5. Orbit A rehabilitated: two stations of one tower

The classification papers set Orbit A aside as "R-flux in name only." Theorem 1 reverses that judgment's valence without touching its content: Orbit A's algebra — obtained directly from $\mathrm{Im}(\mathbb{O})$ when the *coordinate* triple closes — is exactly M_0, the endpoint of the second contraction applied to Orbit B. The full tower reads

$$\mathrm{Im}(\mathbb{O}) \xrightarrow{\text{(1,2,3) contraction, Orbit B}} M_1 \text{ (R-flux Malcev)} \xrightarrow{\text{(1,1,2) contraction}} M_0 \text{ (Heisenberg)} \longrightarrow \text{associative r}$$

Orbit B is the non-associative stratum; Orbit A is the associative-ready quantum-mechanical stratum; the second contraction is the road between them. The $28 + 28$ structure was a map of the tower.

6. The dilation: created by the first contraction, split by the second

Proposition 4 (not a parent symmetry). The grading dilation $\sigma_\lambda : (X, P, W) \mapsto (\lambda X, \lambda^2 P, \lambda^3 W)$ is not an automorphism of $\mathrm{Im}(\mathbb{O})$ for any $\lambda \neq \pm 1$. *Proof.* In the parent, the closed triple satisfies $[P, P] \subseteq P$ with nonzero brackets, forcing $\lambda^2 \cdot \lambda^2 = \lambda^2$. ■

Proposition 5 (μ-uniform automorphism). σ_λ is an automorphism of every M_μ, $\mu \in [0, 1]$: both surviving bracket families are exactly graded (1+1=2, 1+2=3). Verified numerically: automorphism defect 0 (exact) at $\mu \in \{1, 0.3, 0\}$, $\lambda \in \{0.37, 2.5\}$. The dilation symmetry is *created* at the first contraction, at exactly the step where the closed triple's internal bracket — the obstruction in Proposition 4 — is killed.

Proposition 6 (splitting on the fixed-W algebra). On any representation with W frozen to a scalar ($W = \hbar \cdot 1$), σ_λ fails to be an automorphism by exactly the factor λ^3 (verified: the bracket coefficient scales to $\lambda^3 \cdot (-2)$), and it factors uniquely as

$$(\lambda, \lambda^2, \lambda^3) \;=\; \underbrace{(\lambda^{3/2}, \lambda^{3/2}, \lambda^3)}_{\text{uniform: automorphism, rescales } W} \quad\circ\quad \underbrace{(\lambda^{-1/2}, \lambda^{1/2}, 1)}_{\text{squeeze: automorphism, preserves } W},$$

both factors verified as exact automorphisms of their respective structures. The W-preserving squeeze flow, with generator $\frac{1}{2}(XP + PX)$ in any representation, is the component of the octonionic dilation that fixed-$\hbar$ quantum mechanics can carry as an internal symmetry; the W-rescaling component acts across scales and is not an internal symmetry of any fixed-$\hbar$ theory.

7. The SO(3) remnant of G_2

Fixing a grading breaks $G_2 = \mathrm{Aut}(\mathbb{O})$. We compute the unbroken remainder exactly, stating the prediction before the computation, and prove its action.

Derivation. Derivations of a composition algebra are skew with respect to the norm form (Schafer), so a grading-preserving derivation satisfies $D(e_4) \in \mathbb{R}e_4$ together with $\langle D(e_4), e_4 \rangle = 0$, forcing $D(e_4) = 0$. The Leibniz rule then gives $D(e_4 x) = e_4 D(x)$: D commutes with the complex structure $J = L_{e_4}$ on the six-dimensional complement, and preserving the P-triple forces its matrix to be real, hence in $\mathfrak{so}(3)$, acting identically on $X = J(P)$.

Computation (two stages, predictions stated first). Stage 1, the stabilizer of e_4 alone: predicted $\mathfrak{su}(3)$ ($G_2/SU(3) = S^6$, classical); computed dimension: 8. Stage 2, the full grading stabilizer: predicted $\mathfrak{so}(3)$; computed dimension: 3. Per-element verification for all three basis elements: the P-block is antisymmetric; the X-block is *identical* to the P-block; e_4 is annihilated; the three elements close under the bracket.

Proposition 7 (tower survival). The diagonal $SO(3)$ — one rotation R applied simultaneously to the X-triple and the P-triple, with W fixed — is an automorphism of every M_μ. *Proof.* The structure constants of M_μ are built solely from δ_{ab} and ϵ_{abc}, the two $SO(3)$-invariant tensors: invariance under $R^\mathsf{T} R = 1$ and $\det R = 1$ respectively. ■ Verified numerically for a random rotation: automorphism defect 1.3×10^{-15} at both $\mu = 1$ and $\mu = 0$.

Consequence. Within this model, the joint rotational isotropy of three-dimensional position and momentum space — with X and P transforming as vectors under one and the same rotation — is *derived* as the unbroken remnant of the octonionic symmetry, not postulated. The symmetry-breaking chain is $G_2\ (14) \supset \mathfrak{su}(3)\ (8) \supset \mathfrak{so}(3)\ (3)$: fixing the mediator direction leaves $\mathfrak{su}(3)$ (the classical stabilizer of a point on S^6); fixing the further

X/P split leaves exactly the rotations. Eleven generators are spent parametrizing the grading choice, consistent with the orbit dimension $14 - 3 = 11$ and, at the discrete level, with the 28-count of §3.

8. Scope: the program these results prepare

The following is context, stated with citations, not results of this note. The tower of §§4–5 terminates in the CCR structure from which type III_1 factors are built in the operator-algebra literature (Araki–Woods); the type of the factor obtained from any particular state is classified by Connes' invariants, with two independent diagnostics available: the asymptotic ratio set, and the homogeneity of the state space — for type III_1, all normal states are approximately unitarily equivalent (Connes & Størmer, *J. Funct. Anal.* 28 (1978), 187–196). The results above prepare a specific state problem: a state invariant under the $\mathfrak{so}(3)$ remnant of §7 and thermal (KMS) with respect to the squeeze flow of §6. Three structural theorems shape that problem from outside: no infinite-dimensional simple exceptional Jordan algebra exists (Zelmanov), so octonionic content remains finite and local; a JBW algebra is the self-adjoint part of a von Neumann algebra precisely when it admits a dynamical correspondence (Alfsen & Shultz), which the exceptional algebra lacks — observables without time; and a factor is semifinite precisely when its modular flow is inner (Takesaki), so no algebra-resident density can generate the sought flow at the type III target — the state must be characterized by the KMS condition directly. Whether the resulting modular spectrum is the full continuum (type III_1) or a discrete scale lattice (type III_λ) is a computation, carried out in the companion papers on the squeeze-thermal state: either outcome carries the octonionic origin, in the flow or in the scale, and neither is assumed here.

References

- Companion papers: *The Central Element of Algebraic R-Flux*; *Which Contractions of the Octonions Give Genuine R-Flux*; *Malcev-Preservation Is Generic, Not Exceptional*; *The X/P Asymmetry Was Never a Law*.
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